UNIT 3 · TOPIC 3.5

Network Threat Detection

Assignment

Instructions: 17 practice questions + 3 coding assignment(s), 32 points total. Show your work for short-answer items. Coding assignments may be completed here or in the interactive auto-graded environment on the course site (same problem, same points).

Name: **Date:** **Class:**

Practice Questions

3.5.a · Intrusion detection and prevention systems (IDS/IPS)

1. An IDS raises an alert on a packet that turns out to be a legitimate network scanner run by the IT team. This outcome is called a: (1 pt)

- ○ A. True positive
- ○ B. False positive
- ○ C. False negative
- ○ D. True negative

2. Which statement BEST distinguishes an IPS from an IDS? (1 pt)

- ○ A. An IPS uses anomaly detection; an IDS uses signatures.
- ○ B. An IDS monitors host logs; an IPS monitors network traffic.
- ○ C. An IPS can drop malicious packets in real time; an IDS only generates alerts.
- ○ D. An IPS is deployed outside the firewall; an IDS is deployed inside.

3. Select ALL true statements about signature-based detection. (3 pts)

- □ A. It can detect zero-day exploits that have no known signature.
- □ B. It requires the signature database to be kept current.
- □ C. It matches traffic against patterns of known attacks.
- □ D. It typically has a lower false-positive rate than anomaly detection for known attacks.

(Select ALL that apply.)

4. A real attack passes through the IDS without triggering any alert. This is called a: (1 pt)

- ○ A. False negative
- ○ B. False positive
- ○ C. True negative
- ○ D. True positive

5. Name the detection method that builds a behavioral baseline of normal activity and alerts on deviations, even for attacks with no known signature. (1 pt)

3.5.b · Network monitoring and traffic analysis

6. Which data source summarizes a conversation (endpoints, ports, protocol, byte counts) WITHOUT storing packet payload? (1 pt)

- ○ A. A flow record (NetFlow)
- ○ B. Full packet capture (pcap)
- ○ C. A signature database
- ○ D. A connection-tracking table

7. A host normally sends a few MB per day and suddenly uploads several GB to an external server overnight. This is a classic indicator of: (1 pt)

- ○ A. Beaconing
- ○ B. Rule shadowing
- ○ C. Data exfiltration
- ○ D. A false positive

8. Malware checking in with a command-and-control server at exact 60-second intervals is exhibiting: (1 pt)

- ○ A. Baselining
- ○ B. Beaconing
- ○ C. Rule shadowing
- ○ D. A true negative

9. Select ALL indicators that traffic analysis (not signature matching) is well suited to catch. (3 pts)

- □ A. Unusual destination port for a host
- □ B. A known malware file hash
- □ C. A data-volume spike outside business hours
- □ D. Beaconing at regular intervals

(Select ALL that apply.)

10. Name the process of recording a host's typical traffic pattern (volume, ports, timing) so deviations can later be detected. (1 pt)

11. Why do most enterprise monitoring pipelines rely on flow records rather than full packet capture for day-to-day analysis? (1 pt)

- ○ A. Flow records are more secure than packet captures
- ○ B. Flow records are far cheaper to store and query at scale
- ○ C. Packet capture cannot see TCP ports
- ○ D. Flow records include the full payload, which is more useful

3.5.c · Incident response sequence

12. Which incident response phase involves isolating an infected host from the network WITHOUT yet removing the malware? (1 pt)

- ○ A. Identification
- ○ B. Eradication

○ C. Containment

○ D. Recovery

13. Put in order: which phase comes immediately AFTER Containment? (1 pt)

○ A. Eradication

○ B. Identification

○ C. Preparation

○ D. Lessons Learned

14. Select ALL reasons eradicating malware BEFORE containment is a mistake. (3 pts)

□ A. It can tip off the attacker before the host is isolated

□ B. The host may be immediately reinfected from another compromised host on the same network

□ C. It skips the Preparation phase entirely

□ D. It removes the root cause before the spread is stopped

(Select ALL that apply.)

15. A team documents a full incident timeline and updates its response plan afterward. Which phase is this? (1 pt)

○ A. Lessons Learned

○ B. Eradication

○ C. Recovery

○ D. Identification

16. Name the NIST publication that defines the six-phase incident response lifecycle used in this course. (1 pt)

17. Which phase happens continuously BEFORE any incident occurs, and includes writing the IR plan and running tabletop exercises? (1 pt)

○ A. Identification

○ B. Preparation

○ C. Containment

○ D. Recovery

Coding Assignments (Python 3)

3.5.a · Intrusion detection and prevention systems (IDS/IPS)

3 pts · auto-graded on the site

Coding problem 3.5.a: IDS signature matcher

Skill: Detect Attacks · **Maps to:** 3.5.A · **Language:** Python 3 (pytest)

Problem

You are implementing the rule-matching core of a simple IDS. The sensor receives a stream of packet-log entries. Each entry is a dict with the fields shown below. You are given a list of signature rules; each rule has a destination port and a keyword that must appear (case-insensitive) in the payload. A log entry matches a rule when BOTH the port AND the keyword condition are met.

Return the sorted list of `id` values for every log entry that matches at least one rule.

Function contract

Implement `solve(logs: list[dict], rules: list[dict]) -> list[str]` in `starter.py` (see `interface.md` / `blackbox.yaml`).

Input

- `logs`: list of dicts, each with keys: - `"id"` (str), unique entry identifier, e.g. `"pkt-001"` - `"dst_port"` (int), destination port of the packet - `"payload"` (str), packet payload (may be empty)
- `rules`: list of dicts, each with keys: - `"port"` (int), destination port to match - `"keyword"` (str), substring to find (case-insensitive) in the payload

Output

A sorted `list[str]` of `id` values for entries that match at least one rule. Empty list if no entries match.

Examples

```
logs = [
    {"id": "pkt-001", "dst_port": 80,  "payload": "GET /admin/../etc/passwd HTTP/1.1"},
    {"id": "pkt-002", "dst_port": 443, "payload": "TLS handshake"},
    {"id": "pkt-003", "dst_port": 80,  "payload": "GET /index.html HTTP/1.1"},
    {"id": "pkt-004", "dst_port": 22,  "payload": "SSH-2.0-OpenSSH_8.9"},
]
rules = [
    {"port": 80, "keyword": "etc/passwd"},
    {"port": 22, "keyword": "libssh"},
]
solve(logs, rules)  ->  ["pkt-001"]

logs = []
solve(logs, rules)  ->  []
```

Constraints

- Deterministic, offline, standard library only.
- Keyword matching is case-insensitive substring match.
- A log entry that matches multiple rules is included only once in the output.
- Output is sorted ascending by `id`.
- Must pass all cases in `tests.py` (visible + hidden).

How to run

```
cd problem && python -m pytest tests.py -q
```

Starter code:

```
"""Starter code for 3.5.a. Student completes the TODO.
This file intentionally FAILS the tests until implemented."""
from typing import List, Dict, Any


def solve(logs: List[Dict[str, Any]], rules: List[Dict[str, Any]]) -> List[str]:
    """Return sorted list of log-entry IDs whose dst_port and payload match at least one rule.
```

```
    Args:
        logs:  list of dicts with keys 'id' (str), 'dst_port' (int), 'payload' (str)
        rules: list of dicts with keys 'port' (int), 'keyword' (str)
    Returns:
        Sorted list of matching log entry IDs (de-duplicated).
    """
    # TODO: implement per prompt.md
    raise NotImplementedError("implement solve() — see prompt.md")
```

3.5.b · Network monitoring and traffic analysis

3 pts · auto-graded on the site

Coding problem 3.5.b: Traffic-anomaly detector

Skill: Detect Attacks · **Maps to:** 3.5.B · **Language:** Python 3 (pytest)

Problem

Implement a simplified network traffic-anomaly detector. You are given a per-host **baseline** (the normal profile for that host) and a window of **observed flow records**. Flag every host that either exceeds its baseline byte-volume threshold or uses a destination port not in its baseline's allowed-port set.

A **baseline** is a dict mapping host IP (str) to a dict with:

- `"max_bytes"`: integer, the maximum total outbound bytes this host normally sends in the window
- `"allowed_ports"`: list of integers, destination ports this host normally uses

An **observed flow record** is a dict with:

- `"host"`: source host IP string
- `"dst_port"`: destination port integer
- `"bytes"`: integer, bytes sent in this flow

A host may appear in multiple flow records within the same window; volumes accumulate.

Function contract

Implement `solve(baseline, flows)` in `starter.py` (see `interface.md` / `blackbox.yaml`).

- `baseline`: dict as described above
- `flows`: list of flow record dicts (any order)

Return a sorted `list[str]` of host IPs flagged as anomalous. A host is flagged if EITHER: 1. Its **total** bytes across all its flow records in this window exceeds `max_bytes`, OR 2. Any single flow record for that host uses a `dst_port` not present in `allowed_ports`.

A host not present in `baseline` at all is always flagged (unknown host, no baseline to compare against, treated as an anomaly).

Examples

```
baseline = {
    "10.0.1.5": {"max_bytes": 500_000_000, "allowed_ports": [445, 139]},
    "10.0.2.9": {"max_bytes": 5_000_000, "allowed_ports": [80, 443]},
}
flows = [
    {"host": "10.0.1.5", "dst_port": 445, "bytes": 100_000_000},
    {"host": "10.0.1.5", "dst_port": 445, "bytes": 50_000_000},
```

```
    {"host": "10.0.2.9", "dst_port": 4444, "bytes": 1_000},   # unusual port -> flagged
]
solve(baseline, flows) == ["10.0.2.9"]

baseline = {"10.0.3.1": {"max_bytes": 1_000_000, "allowed_ports": [443]}}
flows = [
    {"host": "10.0.3.1", "dst_port": 443, "bytes": 900_000},
    {"host": "10.0.3.1", "dst_port": 443, "bytes": 900_000},   # total 1.8M > 1M -> flagged
]
solve(baseline, flows) == ["10.0.3.1"]
```

Constraints

- Deterministic, offline, standard library only.
- Must pass all cases in `tests.py` (visible + hidden).
- Result is sorted, contains no duplicate host IPs.

How to run

```
cd problem && python -m pytest tests.py -q
```

Starter code:

```
"""Starter code for 3.5.b. Student completes the TODO.
This file intentionally FAILS the tests until implemented.
"""
from typing import Dict, List


def solve(baseline: Dict, flows: List[Dict]) -> List[str]:
    """Flag hosts whose observed traffic deviates from their baseline profile.

    Args:
        baseline: host IP -> {"max_bytes": int, "allowed_ports": list[int]}
        flows:    list of {"host": str, "dst_port": int, "bytes": int}

    Returns:
        Sorted list of anomalous host IPs (no duplicates).
    """
    # TODO: implement per prompt.md
    raise NotImplementedError("implement solve() — see prompt.md")
```

3.5.c · Incident response sequence

3 pts · auto-graded on the site

Coding problem 3.5.c: Incident response phase sorter

Skill: Detect Attacks, Collaborate · **Maps to:** 3.5.C · **Language:** Python 3 (pytest)

Problem

The NIST SP 800-61 incident response lifecycle has six phases, in this canonical order:

```
1. Preparation
2. Identification
3. Containment
4. Eradication
5. Recovery
6. Lessons Learned
```

You will implement a function that takes a list of response-action records, each tagged with the phase it belongs to, but given in shuffled order, and returns the actions sorted into correct lifecycle order. Actions within the same phase keep their original relative order (a stable sort).

An **action record** is a dict with:

- `"action"`: string description of the action
- `"phase"`: string, one of the six canonical phase names exactly as spelled above

Function contract

Implement `solve(actions)` in `starter.py` (see `interface.md` / `blackbox.yaml`).

- `actions`: list of action record dicts, in arbitrary order

Return a `list[str]` of the `"action"` text values, reordered so that all actions from an earlier phase (per the canonical order above) come before all actions from a later phase. Within the same phase, preserve the original input order (stable sort). If an action's `"phase"` value is not one of the six canonical names, raise `ValueError`.

Examples

```
actions = [
    {"action": "Rebuild the server from a clean backup", "phase": "Recovery"},
    {"action": "Write the incident response plan", "phase": "Preparation"},
    {"action": "Isolate the infected host from the network", "phase": "Containment"},
    {"action": "Confirm the IDS alert is a real compromise", "phase": "Identification"},
]
solve(actions) == [
    "Write the incident response plan",
    "Confirm the IDS alert is a real compromise",
    "Isolate the infected host from the network",
    "Rebuild the server from a clean backup",
]
```

```
actions = [
    {"action": "Bogus phase", "phase": "Investigation"},
]
solve(actions)  # raises ValueError
```

Constraints

- Deterministic, offline, standard library only.
- Must pass all cases in `tests.py` (visible + hidden).
- Sort is stable within a phase.

How to run

```
cd problem && python -m pytest tests.py -q
```

Starter code:

```
"""Starter code for 3.5.c. Student completes the TODO.
This file intentionally FAILS the tests until implemented.
"""
from typing import Dict, List

PHASE_ORDER = [
    "Preparation",
    "Identification",
    "Containment",
```

```
    "Eradication",
    "Recovery",
    "Lessons Learned",
]


def solve(actions: List[Dict]) -> List[str]:
    """Sort action records into canonical NIST IR phase order (stable within a phase).

    Args:
        actions: list of {"action": str, "phase": str} dicts, phase must be one of
                 the six canonical NIST SP 800-61 phase names.

    Returns:
        List of "action" strings, reordered by phase.

    Raises:
        ValueError: if any action's phase is not a canonical phase name.
    """
    # TODO: implement per prompt.md
    raise NotImplementedError("implement solve() — see prompt.md")
```